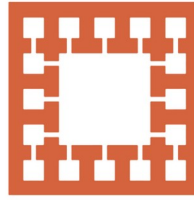


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Varendra University

Dept. of Computer Science & Engineering (CSE)

Lab 1

Course Title: **Microprocessor and Assembly Language Lab**

Course No: **CSE 3206**

Prepared By:
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Lecturer, Dept. of CSE, VU

```
1  #include<stdio.h>
2  int main()
3  {
4      int n;
5      scanf ("%d", &n) ;
6      printf ("%d", n) ;
7      return 0;
8  }
```

Register:

1. Data Register

- I. AX (Accumulator Register)
- II. BX (Base Register)
- III. CX (Count Register)
- IV. DX (Data Register)

- | | | |
|----|---------|----|
| a) | AX = AH | AL |
| b) | BX = BH | BL |
| c) | CX = CH | CL |
| d) | DX = DH | DL |

- a) AX = AH AL
- b) BX = BH BL
- c) CX = CH CL
- d) DX = DH DL

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- AL is reserved for **input**.
- DL is reserved for **output**.

<i>Function number</i>	<i>Routine</i>
1	single-key input
2	single-character output ✓
9	character string output

MOV AH, 1

Opcode

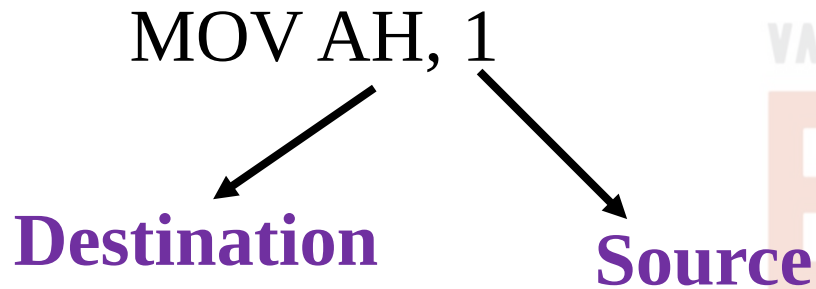
Operand

Opcode: Operation.

Operand: 2 parts,

- Destination.
- Source.

Opcode Destination, Source



Destination: The register or memory location where data is stored.

Source: Provides the data.

MOV: Used to transfer data from one location to another.

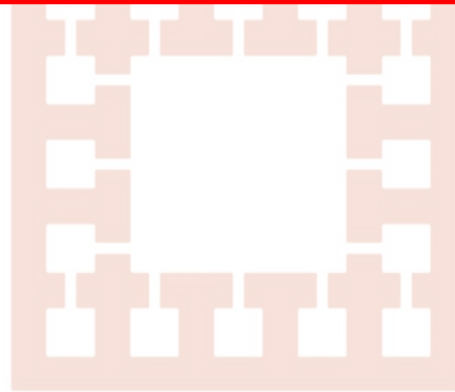
<i>Function number</i>	<i>Routine</i>
1	single-key input
2	single-character output ✓
9	character string output

For Input:

```
mov ah,1  
int 21h  
mov bl,al
```

For Output:

```
mov ah,2  
mov dl,bl  
int 21h
```



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```
01 .model small
02 .stack 100h
03 .code
04
05 main proc
06
07
08     exit:
09     mov ah, 4ch
10     int 21h
11     main endp
12 end main
```



Return

Table 4.4 Memory Models

<i>Model</i>	<i>Description</i>
SMALL	code in one segment data in one segment
MEDIUM	code in more than one segment data in one segment
COMPACT	code in one segment data in more than one segment
LARGE	code in more than one segment data in more than one segment no array larger than 64k bytes
HUGE	code in more than one segment data in more than one segment arrays may be larger than 64k bytes

Experiment No: 01

Name of the Experiment:

- i. Write an assembly language program for simple input and output using EMU8086.
- ii. Write an assembly language program for simple input and output, where the output is displayed on a new line using EMU8086.

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- i. Write an assembly language program for simple input and output using EMU8086.

```
.model small
.stack 100h
.code

main proc
    mov ah,1
    int 21h    ;input
    mov bl,al

    mov ah,2
    mov dl,bl
    int 21h

    exit:
    mov ah,4ch
    int 21h
    main endp
end main
```

ii. Write an assembly language program for simple input and output, where the output is displayed on a new line using EMU8086.

```
.model small
.stack 100h
.code

main proc
    mov ah,1
    int 21h      ;Input
    mov bl,al

    mov ah,2
    mov dl,10
    int 21h
    mov dl,13
    int 21h

    mov ah,2
    mov dl,bl    ;Output
    int 21h

    exit:
    mov ah,4ch
    int 21h
    main endp
end main
```